

Audio-Prompted SAM: Bridging Modality Gap for Audio-Visual Segmentation

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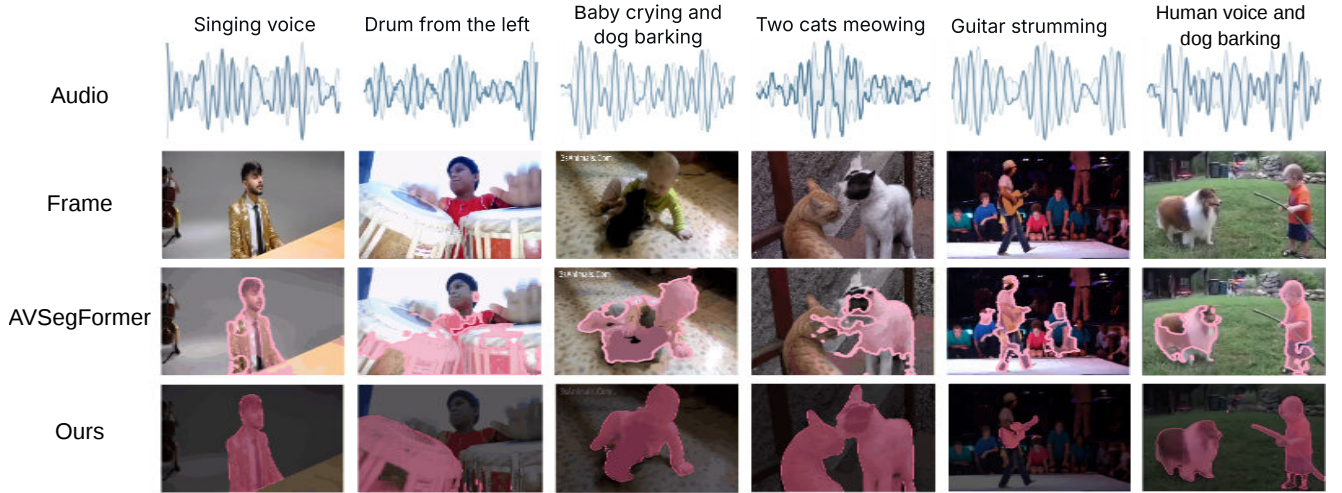


Figure 1. Qualitative comparison of segmentation masks. Each column shows one of four representative examples. AVSegFormer produces fragmented masks with disconnected regions and missing parts due to image feature loss, whereas our method preserves object integrity with complete, connected masks.

Abstract

Audio-Visual Segmentation (AVS) aims to locate and segment sounding objects in video at pixel level. However, existing approaches such as AVSegFormer suffer from poor audio-vision alignment in the embedding spaces and visual information dilution, which leads to wrong and fragmented segmentation mask. To address these challenges, we propose a SAM-based framework (AP-SAM) by treating audio representations as semantic prompts for SAM’s mask decoder. Furthermore, we utilize a contrastive audio-visual alignment loss, and a diversity loss to preserve SAM’s powerful visual priors while explicitly pulling cross-modal embeddings and prevent prompt token collapse. Experiments on the AVSBench MS3 demonstrates +10.96% and +13.5% improvements over AVSegFormer in terms of mIoU and F-score and reduces training time by 75%.

1. Introduction

Audio-Visual Segmentation (AVS) [27] aims to locate and segment sounding objects in a video frame using both visual and audio information. Unlike visual-based segmentation, AVS uses extra cues from audio when the visual features are inefficient. However, matching an audio signal with its correct visual source is challenging. Due to their different feature dimensions, learning reliable cross-modal correspondences becomes difficult.

Reliable AVS is important for many real-world applications, including robotics [28], human-computer interaction [15] and intelligent surveillance [14]. Unlike vision-based approaches, AVS can locate the object using audio signals to make it more effective for complex situations.

Recent AVS methods, such as AVSegFormer, use transformer-based modules to learn audio and visual representations [6]. Despite these advances, there are still limitations: (1) Audio and visual features are often located in different semantic spaces, making feature extraction diffi-

[†]Provide high-level, hands-off supervision and feedback.

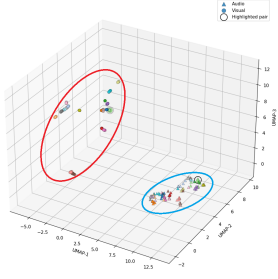


Figure 2. A misalignment using UMAP projection of audio and visual embeddings from AVSegFormer on the MS3 validation set

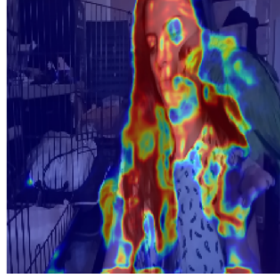


Figure 3. Mask output of AVSegFormer on the MS3 validation set without binarization, showing a fragmented image features.

cul as shown in Fig. 2. (2) The predicted mask values are highly fragmented and irregularly distributed, failing to delineate a cohesive object structure. This occurs because the segmentation head is trained in isolation from the visual encoder, where audio and visual features are naively concatenated along the channel dimension [7]. These shortcomings motivate us to explore a more suitable architecture that preserves visual representations while promoting cross-modal interaction.

The Segment Anything Model (SAM) [12] is a foundational vision model designed for general visual segmentation with remarkable zero-shot generalization. It has a remarkable ability for visual segmentation. However, even with such a powerful backbone, directly combining audio and visual features still suffers from cross-modal semantic misalignment, limiting segmentation performance in sound-guided scenarios. To address this, we propose to treat audio representations as semantic prompts that guide SAM’s mask decoder. Furthermore, we introduce new loss functions to improve semantic alignment between modalities and encourage diverse audio representations. As shown in Fig. 1 and Tab. 1, our approach achieves consistent improvements over strong baselines, demonstrating the effectiveness of prompt-based guidance and the proposed losses.

Our contributions are summarized as follows:

- (1) We proposed a SAM-based generation method to replace the original segmentation structure in AVSegFormer, which better retains the feature of input frame.
- (2) We adopted a prompt-guided generation structure to better balance image features and audio guidance.
- (3) Introduce a CLIP-based alignment loss to improve the feature fusion and diversity loss to prevent the collapse of audio features.
- (4) Demonstrate improved AVS performance in benchmark datasets, improving the mIoU by 10.96% over AVSegFormer on the MS3.

2. Multi-modality pitfalls

AVSegFormer [6] operates in four stages: (1) An audio encoder processes log-mel spectrograms into audio features; (2) A ResNet backbone extracts visual features; (3) A cross-attention transformer fuses the two modalities; (4) A lightweight decoder produces pixel-wise masks.

The key limitation is that fusion occurs in a learned space where audio and visual representations are not semantically aligned. Also, as noted by Han et al. [7], the image features are not directly transmitted to the decoding layer, which degrades mask quality. We visualized this phenomenon in Fig. 3, where the decoder without direct visual feature access produces noticeably blurrier object boundaries.

While AVSegFormer [6] significantly improves the accuracy of the audio-visual segmentation, there are still some problems that prevent higher performance.

2.1. Audio-Visual Misalignment

A key limitation of AVSegFormer is that its audio and visual features reside in poorly aligned embedding spaces. The audio encoder and visual encoder are independently on distinct tasks, resulting in representations that lack shared semantic structure. When these features are fused via cross-attention, the model must learn cross-modal correspondences from a limited AVS dataset, which is insufficient to bridge the modality gap.

Fig. 5 visualizes this misalignment using UMAP [21] projection of audio and visual embeddings from AVSegFormer on the MS3 validation set. The audio features (triangle) and visual features (circle) form two largely separated clusters, confirming that the two modalities occupy distinct regions in the embedding space. This separation explains why direct feature fusion produces suboptimal results.

The misalignment is a problem in multi-source scenarios (MS3) where multiple sounding objects are present. Without proper alignment, the model cannot reliably determine which visual region corresponds to which audio source.

2.2. Loss of Visual Characteristics

Beside misalignment, AVSegFormer suffers from a loss of visual characteristics. In AVSegFormer, the audio and visual features are concatenated along the channel dimension before being passed to the decoder. This operation dilutes the original visual information by mixing it with audio features that originate from a different semantic space [7].

To illustrate this, we analyze the non-binarized segmentation outputs directly generated by the decoder. We observe that the predicted mask values are highly fragmented and irregularly distributed, failing to delineate a cohesive object structure. Although the model still maintains an acceptable mIoU, this severe spatial fragmentation explicitly demonstrates that the original visual features are fundamentally lost during the concatenation.

Consequently, the fused representation makes the model struggle with boundary segmentation. The resulting fragmented output serves as direct evidence of a fundamental architectural bottleneck that cannot be remedied by simply adjusting the fusion mechanism.

3. Method

Notations. We denote input video frames and audio spectrograms as $\mathbf{x}_{vis}$ and $\mathbf{x}_{audio}$, encoded by E_{vis} and E_{audio} into visual embedding $\mathbf{V}$ and audio prompt tokens P_a . The model predicts binary mask $\mathbf{M}_{pred}$ and reconstructed spectrogram $\mathbf{S}_{pred}$, with ground-truth $\mathbf{M}_{gt}$ and target $\mathbf{S}_{target}$. Pooled embeddings P_a and $\bar{\mathbf{V}}_{fg}$ are used for cross-modal alignment via similarity matrices $\mathbf{S}_{a2v}$, $\mathbf{S}_{v2a}$ with temperature τ . The overall loss combines segmentation, reconstruction, alignment, and diversity terms with balancing weights λ and margin m . See Tab. 4 for complete summary.

3.1. Design Motivation

The empirical analysis identifies two fundamental limitations of AVSegFormer: (1) audio-visual semantic misalignment, where the two modalities occupy different embedding spaces; (2) loss of visual characteristics, where feature fusion dilutes visual information, leading to fragmented mask predictions.

To address these problems, we propose a SAM-based framework with three key design choices:

1. A prompt-guided architecture that avoids direct feature fusion, preserving SAM’s visual priors while using audio as conditional prompts;
2. A contrastive alignment loss that explicitly pulls audio and visual embeddings closer, targeting the misalignment documented in Fig. 2;
3. A diversity loss that prevents audio prompt collapse, ensuring rich and informative conditioning signals.

3.2. Overall Architecture

Our proposed framework AP-SAM restructures this pipeline by leveraging SAM’s frozen visual encoder and using audio-based prompt to guide the mask generation.

Fig. 7 illustrates the overall architecture. In contrast to previous AVSegFormer methods [6], we use a prompt-guided mask generation instead of conventional decoder. Specifically, we design the prompt generator by cross-attention machine and add contrastive learning loss to align visual features with audio. Besides, audio diversity loss can prevent the collapse of audio features.

3.3. Architecture Components

Our framework comprises four key components: (1) a frozen SAM image encoder that preserves visual features; (2) a VGGish audio encoder that extracts acoustic features;

(3) a prompt generator with Q-former and cross-attention that transforms audio into semantic prompts; (4) SAM’s mask decoder that produces the final segmentation with prompts.

3.3.1. Vision Encoder

We use SAM’s frozen image encoder (ViT-H [3]) to extract high-quality visual embeddings. The encoder is pre-trained to produce features that are not only semantically rich but also spatially sensitive to object boundaries and edges.

3.3.2. Audio Encoder

We use the VGGish CNN [8] backbone to process log-mel spectrograms to extract abstract audio features. The encoder outputs a fixed-dimensional audio embedding per frame, providing a feature embedding of the input signal.

3.3.3. Prompt Generator

We design an audio-visual prompt generator that transforms audio features into prompts for SAM. The generator consists of a Q-Former [16] with learnable query tokens, and cross-attention with frozen VGGish embeddings through transformer blocks, selecting task-relevant audio semantics into query representations. Then, we use a linear projection $\mathbf{W}_{proj}$ in $\mathbb{R}^{d \times d}$ to map the output of the audio encoder $E_{audio}(\mathbf{x}_{audio})$ to SAM’s prompt embedding space, enabling cross-modal interaction without updating either the audio backbone or the visual encoder.

$$P_a = \mathbf{W}_{proj}(\text{Q-Former}(E_{audio}(\mathbf{x}_{audio}))) \quad (1)$$

E_{audio} is the VGGish audio encoder, Q-Former is the query transformer, and $\mathbf{W}_{proj}$ projects to SAM’s prompt space. Only the prompt generation network (Q-Former + projection) and optionally SAM’s mask decoder are trainable.

This design has these main advantages:

1. **Preserved the original visual features.** SAM’s image encoder features are not corrupted by audio features.
2. **Reduced modality interference.** Audio affects mask generation through attention modulation in the mask decoder rather than through additive feature fusion.

3.3.4. Mask Decoder

The projected audio prompt tokens are used in SAM’s mask decoder as prompt embeddings, where they interact with the image encoder’s visual features. The decoder then integrates these prompts with the image embeddings to produce a segmentation mask, leveraging SAM’s pre-trained ability to generate high-quality masks from audio cues.

3.4. Loss Function

The total training objective is a weighted combination of four terms:

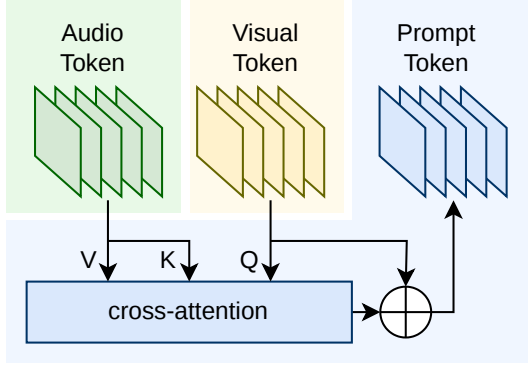


Figure 4. Architecture of the prompt generator. The Visual Tokens serve as the Query, while the Audio Tokens form the Key and Value. Through cross-attention, the model learns task-relevant multimodal information and outputs prompt embeddings to guide the decoder masking.

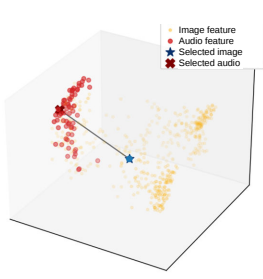


Figure 5. Audio-visual embedding space **before** alignment. Two modalities form largely separated clusters.

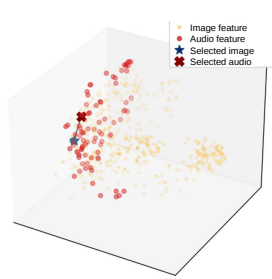


Figure 6. Audio-visual embedding space **after** applying the proposed alignment loss. The gap narrows but persists.

$$L = L_{seg} + \lambda_1 L_{audio} + \lambda_2 L_{align} + \lambda_3 L_{div} \quad (2)$$

where λ denotes the loss-weighting hyperparameter, with λ_1 , λ_2 , and λ_3 controlling the contribution of each auxiliary loss term.

3.4.1. Segmentation Loss

Standard binary segmentation loss using Dice loss:

$$L_{seg} = L_{Dice}(\mathbf{M}_{pred}, \mathbf{M}_{gt}), \quad (3)$$

where $\mathbf{M}$ denotes the segmentation mask, where each pixel value indicates the probability of belonging to the target object. $\mathbf{M}_{pred}$ is the predicted mask and $\mathbf{M}_{gt}$ is the ground-truth mask.

3.4.2. Audio Reconstruction Loss

In addition to the segmentation loss and alignment loss, we introduce an auxiliary audio reconstruction loss to provide a weakly supervised signal for the audio encoder:

$$L_{audio} = \|\mathbf{S}_{pred} - \mathbf{S}_{target}\|_1 \quad (4)$$

Besides, $\mathbf{S}_{pred}$ is the predicted separated spectrogram, $\mathbf{S}_{target}$ represents the target sound source’s ground-truth spectrogram. This design concept was inspired by the work of SepFormer [23], which demonstrated the superiority of the Transformer architecture in speech separation tasks. This L1 loss forces the model to separate the spectral characteristics of the target sound source from the mixed audio. Through the reconstruction task at the spectrogram level, the audio encoder is forced to learn to distinguish sound sources that should be segmented from background noise, thereby generating cleaner and more discriminative audio features for the Q-Former to use.

3.4.3. Audio-Visual Alignment Loss

Our primary contribution for cross-modal alignment. Fig. 5 visualizes the embedding space before training, where audio and visual features form separated clusters. After applying the alignment loss, Fig. 6 shows the two features converging while preserving their respective internal structures, demonstrating that the alignment loss effectively bridges the cross-modal gap. This loss operates on SAM prompt embeddings and visual encoder features with two components:

1. Cosine alignment pulls each audio prompt summary toward visual features:

$$L_{cos} = 1 - \cos(\bar{\mathbf{P}}_a, \bar{\mathbf{V}}_{fg}) \quad (5)$$

2. InfoNCE [24] alignment treats other batch samples as negatives:

$$L_{contra} = \frac{1}{2} [\text{CE}(\mathbf{S}_{a2v}/\tau) + \text{CE}(\mathbf{S}_{v2a}/\tau)], \quad (6)$$

where we define Sim_{a2v} as the similarity matrix from audio to vision.

3.4.4. Diversity Loss

To prevent audio prompt tokens from collapsing to similar representations, we apply a diversity loss that pushes different audio embeddings apart from one another.

$$L_{div} = \sum_{i \neq j} \max(0, m - \|\mathbf{p}_i - \mathbf{p}_j\|) \quad (7)$$

where $\mathbf{p}_i$ is the per-sample audio summary vector obtained by mean-pooling the N_q query tokens from Q-Former followed by ℓ_2 normalization.

4. Related Works

4.1. Audio-Visual Segmentation

In 2022, AVSBench [27] introduced the AVS task along with benchmark datasets containing single-source (S4) and

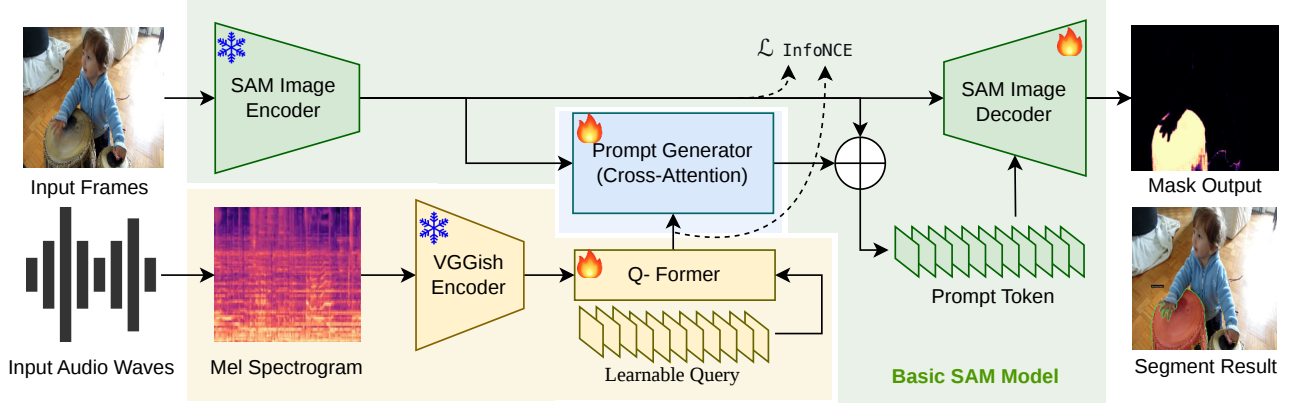


Figure 7. Overall architecture of AP-SAM. Four key components in our framework: (1) Basic SAM pipeline; (2) An AVSegFormer-style audio encoder. (3) Prompt generator using cross-attention layer. (4) The SAM decoder with audio-based prompt.

multi-source (MS3) scenarios. Existing approaches can be broadly categorized by their fusion strategy.

A year later, CATR [17] employs cross-modal transformer layers to exchange information between modalities.

In 2024, AVSegFormer [6] uses a cross-attention mechanism for audio-visual feature interaction, followed by a transformer-based decoder for mask prediction.

Other recent works explored adapting local context [28] or bridging modalities through pretrained models [14].

A current challenge across these approaches is that audio and visual features originate from independently pretrained encoders and located in different embedding spaces. Direct feature fusion introduces noise and limits segmentation quality.

4.2. Prompt Learning

Prompt learning has been used for adapting large pretrained models to different tasks [11]. For SAM [12], promptable segmentation uses sparse prompts like points, boxes and dense prompts like masks to condition mask generation.

Recent work explores prompting across modalities. In multimodal tasks, prompts can carry information from one modality to guide another. We extend this idea to AVS by using audio-based prompts. The audio signal is encoded into a set of learnable query tokens that serve as prompts for the mask decoder.

5. Experimental Setup

5.1. Dataset

We evaluate on the MS3 subset of AVSBench [27], which contains videos with multiple sounding objects. The dataset consists of 424 videos from 25 categories, split into 298 training, 48 validation, and 78 testing videos following the standard AVSegFormer partition. Each video is approxi-

mately 5 seconds long at 24 fps, yielding 120 frames per video.

For preprocessing, we follow the protocol established by AVSegFormer [6]. Visual frames are resized to 512×512 when using SAM’s image encoder and normalized using ImageNet statistics. Audio signals are converted to log-mel spectrograms with 256 frequency bins using a Hann window of 25 ms and hop length of 10 ms. Data augmentation includes random horizontal flip and random brightness/contrast jitter for visual frames.

5.2. Implementation Details

Model Configuration. We use SAM ViT-H [12] as our frozen visual encoder. The audio encoder is a VGGish-style CNN with 4 convolutional blocks and global pooling, producing 128-dim audio embeddings. The Q-Former consists of 4 transformer layers with 8 attention heads and 32 learnable query tokens (256-dim each). The projection layer is a single linear layer mapping from 256-dim to SAM’s prompt embedding dimension.

Training Setup. We use the AdamW [18] optimizer with learning rate $1e-4$, and batch size 10 (per GPU). Training runs for 30 epochs. The loss weights in Eq. (2) are set to $\lambda_1 = 0.1$, $\lambda_2 = 0.1$, $\lambda_3 = 0.05$. The contrastive temperature $\tau = 0.07$ and diversity margin $m = 0.5$.

Hardware. All experiments are conducted on a single NVIDIA GPU (40 GB memory, A100) with PyTorch 2.5.1. The SAM image encoder forward pass is the main computational bottleneck but remains frozen during training.

5.3. Evaluation Metrics

We adopt two standard metrics for AVS evaluation:

Mean Intersection over Union (mIoU). mIoU com-

putes the average IoU across all test samples:

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^N \frac{|\mathbf{M}_{pred} \cap \mathbf{M}_{gt}|}{|\mathbf{M}_{pred} \cup \mathbf{M}_{gt}|} \quad (8)$$

is where $\mathbf{M}_{pred}$ and $\mathbf{M}_{gt}$ are the predicted and true binary masks, respectively.

F-score. This coefficient measures the overlap between prediction and ground truth:

$$F = \frac{1}{N} \sum_{i=1}^N \frac{2 \cdot TP^{(i)}}{2 \cdot TP^{(i)} + FP^{(i)} + FN^{(i)}}$$

Following AVSegFormer, we also report per-category mIoU to provide fine-grained analysis of model performance across different object types.

6. Results

6.1. Quantitative Comparison

We compare AP-SAM with AVSegFormer [6] on the MS3 benchmark. As shown in Tab. 1, our proposed framework achieves improvements across both metrics.

AP-SAM achieves a +10.96% improvement in mIoU and +13.5% in F-score over the AVSegFormer baseline. The improvement validates the effectiveness of our SAM-based architecture with audio prompt conditioning and the proposed alignment loss.

6.2. Qualitative Visualization

We provide qualitative analysis from the perspective of image feature loss. Fig. 1 presents four sets of comparative results, where each set displays (from top to bottom): the AVSegFormer prediction, and our methods.

AVSegFormer relies on a learnable decoder to decode masks directly from feature maps, lacking prior features of the integrity of image features. Consequently, its predictions exhibit severe fragmentation in regions like object boundaries, occluded areas, and low-contrast regions. Specifically, the same object is split into multiple disconnected regions and holes or missing parts in regions with weak feature responses. These issues are a direct manifestation of the image feature loss problem analyzed in §2.

In contrast, benefiting from SAM’s visual priors, AP-SAM predicts more complete masks that preserve object integrity under the same conditions. Even in regions with severe image feature loss, SAM’s visual priors compensate for the missing information and recover complete object boundaries.

6.3. Effect of Proposed Loss and Ablation Study

We conduct ablation studies to evaluate the contribution of each component in our proposed framework. The results are summarized in Tab. 2.

When we remove the audio-visual alignment loss, the model relies on the Q-Former to extract audio information, without explicit cross-modal semantic alignment. In this setting, performance drops to 0.4393 mIoU, which is even lower than the AVSegFormer baseline (0.5146). This confirms that without semantic alignment, the audio prompts are not properly grounded in the visual feature space, causing the SAM mask decoder to receive misleading signals.

Incorporating the alignment loss significantly boosts performance to 0.5255 mIoU. This dramatic improvement highlights that cross-modal alignment is the critical bottleneck in our framework. The alignment loss effectively pulls audio and visual features closer in the shared embedding space, as confirmed by the UMAP visualization (Fig. 6 vs Fig. 5).

We expect the diversity loss to further improve performance by preventing audio prompt token collapse, ensuring that each query token captures distinct semantic information. This should lead to more informative prompt representations and improved mask quality, particularly in multi-source scenarios.

6.4. Efficiency Analysis

We analyze the computational efficiency of AP-SAM in terms of model parameters and training time.

AP-SAM consists of three components: (1) SAM ViT-H image encoder, (2) VGGish audio encoder, (3) Q-Former + projection. Total trainable parameters are under 10M, which is substantially smaller than the full AVSegFormer model that requires updating all parameters.

Training requires 27.8 minutes on a single A100 GPU for 30 epochs. The majority of training time is spent on the SAM image encoder forward pass. For comparison, AVSegFormer’s training takes 92.0 minutes under the same hardware conditions.

AP-SAM offers a favorable trade-off: performance improvement (+10.96 mIoU) with a decrease in training cost, thanks to the frozen SAM backbone and efficient prompt-based design.

7. Limitation and Conclusion

Limitations Despite the improvements, several limitations remain:

1. The audio prompts become less reliable in environments with multiple concurrent sound sources, extreme background noise, or non-stationary audio events. The VGGish encoder’s fixed representations may not generalize well to acoustic conditions outside the pretraining distribution, and the linear projection to SAM’s prompt space may be too limited to capture complex audio semantics.
2. As shown by the UMAP visualization (Fig. 6), our alignment loss reduces but does not eliminate the modality

Table 1. Quantitative comparison on MS3 dataset [27]. We report the mean Intersection over Union (mIoU) and F-score across all methods. Our approach, achieves a performance gain over the previous best method (AVSegFormer) by 10.96% in mIoU and 13.5% in F-score.

Method	Backbone	mIoU	F-score	Reference
LVS [1]	ResNet-50	29.45	33.0	CVPR’2021
MSSL [22]	ResNet-18	26.13	36.3	ECCV’2020
3DC [19]	ResNet-34	36.92	50.3	BMVC’2020
SST [4]	ResNet-101	42.57	57.2	CVPR’2021
iGAN [20]	Swin-T	42.89	54.4	ArXiv’2021
LGVT [26]	Swin-T	40.71	59.3	NeurIPS’2021
AVSBench [27]	ResNet-50	47.88	57.8	ECCV’2022
AVSegFormer [6]	ResNet-50	51.46	63.3	AAAI’2024
AP-SAM (Ours)	ViT-Huge	62.42 (+10.96%)	76.9 (+13.5%)	

Table 2. Ablation study: effect of proposed loss components. We evaluate the individual and joint contributions of the proposed Alignment loss and Diversity loss on the MS3 validation set. The results demonstrate that applying both losses together yields the optimal performance, confirming that enforcing feature alignment while penalizing redundancy is crucial for accurate audio-visual segmentation.

Setting	mIoU	F-score
Proposed model only (no alignment, no diversity)	0.4393	0.5321
Proposed model + Alignment loss	0.5255	0.6388
Proposed model + Alignment loss + Diversity loss	0.6242	0.7687

Table 3. Model size comparison of our method.

Component	Parameters	Status
SAM ViT-H (image encoder)	637.20M	Frozen
VGGish (audio encoder)	4.86M	Frozen
Q-Former + Decoder	8.37M	Trainable
Total trainable parameters	8.37M	
Total parameters	650.43M	

gap. This residual misalignment may fundamentally cap further improvements.

3. We evaluate only on the MS3 subset of AVSBench. Generalization to other AVS datasets and to related tasks remains untested. Future work should expand evaluation to establish broader applicability.

Conclusion In this paper, we addressed the challenges of audio-visual semantic misalignment and visual information dilution in existing audio-visual segmentation methods.

We proposed AP-SAM, a SAM-based framework that utilizes audio representations as semantic prompts for the mask decoder, together with contrastive audio-visual alignment and diversity losses to achieve effective cross-modal conditioning while preserving SAM’s visual priors.

Experiments on the AVSBench MS3 benchmark demonstrate that AP-SAM achieves significant improvements over existing audio-visual segmentation methods.

Future Work Building upon the effectiveness of AP-SAM, several promising directions remain for future exploration.

1. Future work will investigate scaling AP-SAM to larger-scale audio-visual segmentation benchmarks, such as AVSS. To improve the efficiency of model adaptation, parameter-efficient fine-tuning strategies, including LoRA [10] and adapter-based tuning [9], can be incorporated to reduce training overhead while maintaining the capability of SAM-based segmentation.
2. Improving cross-modal representation alignment is another important direction. Future studies could explore more advanced alignment objectives, including SimCLR-style contrastive learning [2] and geodesic contrastive learning [25], to better model the relationship between audio and visual modalities. Moreover, knowledge distillation from large-scale pretrained audio encoders, such as CLAP [5], provides a promising approach to further enhance audio representations and alleviate the modality gap in audio-visual segmentation.
3. Another potential direction is adaptive audio prompt learning. Inspired by Matryoshka Representation Learning [13], future work could apply nested representation constraints to learnable audio prompt tokens, enabling the model to dynamically utilize different numbers of tokens according to input complexity. Such a sparse and adaptive prompting mechanism may improve both efficiency and robustness for diverse audio-visual scenes.

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Audio-Prompted SAM: Bridging Modality Gap for Audio-Visual Segmentation

Supplementary Material

Table 4. Notation summary

Symbol	Description
<i>Inputs & Encoders</i>	
$\mathbf{x}_{\text{vis}}, \mathbf{x}_{\text{audio}}$	Input video frame and audio spectrogram
$E_{\text{vis}}(\cdot), E_{\text{audio}}(\cdot)$	SAM image encoder, VGGish audio encoder
$\mathbf{V}$	Visual embedding from SAM encoder
$\mathbf{W}_{\text{proj}}$	Linear projection to SAM prompt space
$P_a, \mathbf{p}_i$	Audio prompt tokens; i -th token
<i>Masks & Spectrograms</i>	
$\mathbf{M}_{\text{pred}}, \mathbf{M}_{\text{gt}}$	Predicted and ground-truth binary masks
$\mathbf{S}_{\text{pred}}, \mathbf{S}_{\text{target}}$	Predicted and target audio spectrograms
<i>Alignment</i>	
$\bar{P}_a, \bar{\mathbf{V}}_{fg}$	Pooled audio prompt and foreground visual embeddings
$\mathbf{S}_{a2v}, \mathbf{S}_{v2a}$	Audio-to-visual and visual-to-audio similarity matrices
τ	InfoNCE temperature
<i>Losses</i>	
$L_{\text{seg}}, L_{\text{audio}}, L_{\text{align}}, L_{\text{div}}$	Segmentation, reconstruction, alignment, diversity losses
$\lambda_1, \lambda_2, \lambda_3$	Weights for $L_{\text{audio}}, L_{\text{align}}, L_{\text{div}}$
m	Diversity loss margin